

Comparing Java and Python

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Having knowledge of multiple programming languages is critical in the field of software development. Just as a contractor might have a screwdriver, a drill, and a hammer, a programmer uses programming languages as tools, and knowing when to use each one is important for building the best possible solution. This journal compares the history, characteristics, and usefulness of two object-oriented programming languages: Java and Python. Learning multiple programming languages expands a developer's toolset, strengthens problem-solving abilities, deepens the understanding of programming concepts, and improves debugging skills (Kumar, 2025).

Comparison of Java and Python

While both Python and Java are object-oriented programming languages, each has unique characteristics that make them valuable for different types of software development. According to Aggarwal (2025), Python is an interpreted language, whereas Java is a compiled language. This means Python executes code line by line during runtime, while Java compiles code into bytecode before it is executed by the Java Virtual Machine (JVM).

Python also typically requires fewer lines of code to accomplish the same task, making it easier to read and faster to develop with. Java generally requires more code, but this additional structure contributes to improved performance and reliability. Java is statically typed, meaning variable types are declared before execution, while Python is dynamically typed and determines data types during runtime. Although Python's flexibility speeds up development, Java's static typing can help catch programming errors earlier in the development process.

The two languages are also commonly used for various applications. Python is widely used for artificial intelligence, machine learning, data science, automation, and web development because of its simple syntax and extensive library ecosystem. Java is commonly used for enterprise software, Android applications, financial systems, desktop applications, and large-scale web services because of its performance, scalability, and platform independence (Aggarwal, 2025).

When selecting a programming language, several factors should be considered, including the target platform, performance requirements, available libraries, maintainability, and the project's overall goals. For projects that require rapid development and readable code, Python is often an excellent choice. For applications that demand high performance, strong database connectivity, and cross-platform compatibility, Java is often the better option because it follows the “Write Once, Run Anywhere” (WORA) principle (Aggarwal, 2025).

History of Java and Python

Sun Microsystems originally developed Java as part of its Green Project. It was publicly released in 1995 and has since become one of the most widely used programming languages for enterprise, web, mobile, and cloud-based applications. What began with only a few hundred classes has grown into thousands of APIs that support a wide range of software systems. Today, Java is widely used in secure enterprise applications, desktop software, web services, cloud computing, and big data technologies for its reliability, scalability, and platform independence (Bhatnagar, 2026).

Python was first released in 1991 by Guido van Rossum and was inspired by the comedy series *Monty Python's Flying Circus*. Unlike Java, developed by a large technology company,

Python began as the vision of a single developer seeking to make programming more intuitive and enjoyable. Since then, Python has grown into one of the fastest-growing programming languages and is now maintained by the Python Software Foundation (PSF). Its readability, versatility, and ease of use have made it especially popular for artificial intelligence, data science, automation, web development, and education (Python Institute, n.d.).

Although Java and Python were created with different goals in mind, both languages have continuously evolved to meet the changing needs of software development. Java has remained a leading choice for large-scale enterprise systems because of its performance and stability, while Python has become a preferred language for emerging technologies such as artificial intelligence and machine learning due to its simplicity and extensive ecosystem of libraries. Their continued growth demonstrates how programming languages can adapt over time while remaining relevant in modern software development.

Usefulness of Java and Python

Although Java and Python were developed more than 30 years ago, both languages continue to play significant roles in modern software development. Rather than one language replacing the other, each has found its own niche based on its strengths. As technology has evolved, both languages have continued to receive updates, new libraries, and community support, keeping them relevant across a wide variety of industries.

Java remains one of the most common choices for enterprise software because of its performance, reliability, and security. It is frequently used to develop large-scale business applications, banking systems, cloud services, Android applications, and backend web services where long-term stability and scalability are essential (Bhatnagar, 2026). Its platform

independence also allows developers to write software once and deploy it across multiple operating systems with minimal changes.

Python has become one of the fastest-growing programming languages because it emphasizes readability and rapid development. It is widely used in artificial intelligence, machine learning, automation, data science, scientific computing, and web development. Its straightforward syntax allows developers to build and test applications quickly while leveraging an extensive collection of community-supported libraries (Python Institute, n.d.). As fields such as artificial intelligence and automation continue to grow, Python's importance is likely to continue increasing alongside them.

Benefits of Knowing Multiple Languages

Learning multiple programming languages allows developers to become more adaptable and better prepared for a wide variety of programming challenges. Since every language has its own strengths and weaknesses, understanding more than one language makes it easier to choose the best tool for a particular project. It also exposes programmers to different approaches for solving problems, leading to a deeper understanding of programming concepts rather than simply memorizing the syntax of a single language (Kumar, 2025).

Knowing multiple languages also makes learning additional languages easier because many programming concepts carry over from one language to another. Variables, loops, conditionals, functions, classes, and object-oriented principles are found in many modern programming languages, even when the syntax differs. As developers become familiar with these core concepts, they can spend less time learning the fundamentals and more time understanding the unique features of a new language. According to Kumar (2025), learning multiple

programming languages also improves debugging skills, increases adaptability, and broadens career opportunities by enabling developers to contribute to a wider variety of projects.

Programming languages should not be viewed as competitors but as tools designed for different purposes. Python's simplicity makes it an excellent choice for rapid development, automation, and data-driven applications, while Java provides the structure, scalability, and performance required for many enterprise systems. Understanding both languages gives developers greater flexibility when selecting the appropriate technology for a project and strengthens the foundational knowledge needed to learn additional languages in the future.

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